HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2012)

Student No.: 05 Name: _____ Marks: / 110

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. $(4x+y)^2-(4x-y)^2=$

A. 0.

B. $2y^2$.

C. $8xy$.

D. $16xy$.

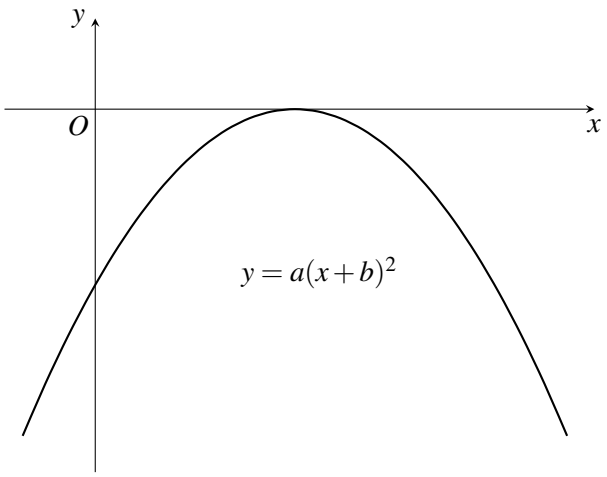
(DSE-2012 PII Q02)

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2. The figure shows the graph of $y = a(x+b)^2$, where $a$ and $b$ are constants. Which of the following is true?




A. $a > 0$ and $b > 0$

B. $a > 0$ and $b < 0$

C. $a < 0$ and $b > 0$

D. $a < 0$ and $b < 0$

(DSE-2012 PII Q06)

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3. The solution of $15+4x<3$ or $9-2x>1$ is

A. $x<-3$.

B. $x>-3$.

C. $x<4$.

D. $x>4$.

(DSE-2012 PII Q07)

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4. In a company, 37.5% of the employees are female. If 60% of the male employees and 80% of the female employees are married, then the percentage of married employees in the company is

A. 32.5%.

B. 45%.

C. 55%.

D. 67.5%.

(DSE-2012 PII Q08)

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5. Mary performs a typing task for 7 hours. Her average typing speeds for the first 3 hours and the last 4 hours are 63 words per minute and 56 words per minute respectively. Find her average typing speed for the 7 hours.

A. 17 words per minute

B. 35 words per minute

C. 59 words per minute

D. 60 words per minute

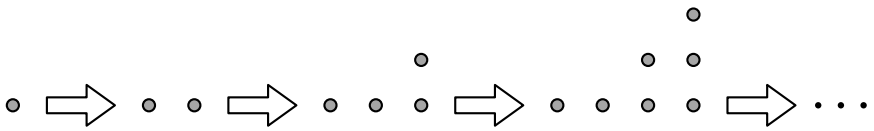
(DSE-2012 PII Q11)

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6. In the figure, the 1st pattern consists of 1 dot. For any positive integer $n$, the $(n+1)$th pattern is formed by adding $n$ dots to the $n$th pattern. Find the number of dots in the 8th pattern.



A. 22

B. 29

C. 36

D. 37

(DSE-2012 PII Q12)

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7. The length of a piece of thin string is measured as 25 m correct to the nearest m. If the string is cut into $n$ pieces such that the length of each piece is measured as 5 cm correct to the nearest cm, find the greatest possible value of $n$.

A. 445

B. 566

C. 567

D. 650

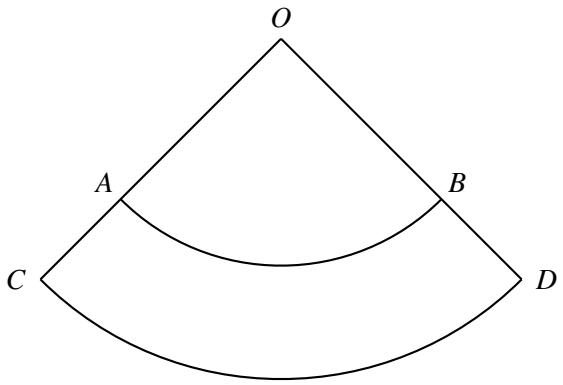
(DSE-2012 PII Q14)

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8. In the figure, $OAB$ and $OCD$ are sectors with centre $O$. If arc $AB = 12\pi$ cm, arc $CD = 16\pi$ cm and $OA = 30$ cm, then $AC =$




A. 5 cm.

B. 10 cm.

C. 20 cm.

D. 40 cm.

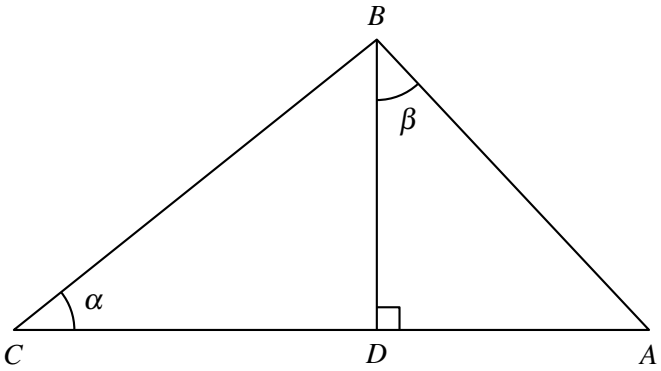
(DSE-2012 PII Q16)

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9. In the figure, $D$ is a point lying on $AC$ such that $BD$ is perpendicular to $AC$. If $BC = \ell$, then $AB =$




A. $\dfrac{\ell \sin \alpha}{\cos \beta}$.

B. $\dfrac{\ell \sin \beta}{\cos \alpha}$.

C. $\dfrac{\ell \cos \alpha}{\sin \beta}$.

D. $\dfrac{\ell \cos \beta}{\sin \alpha}$.

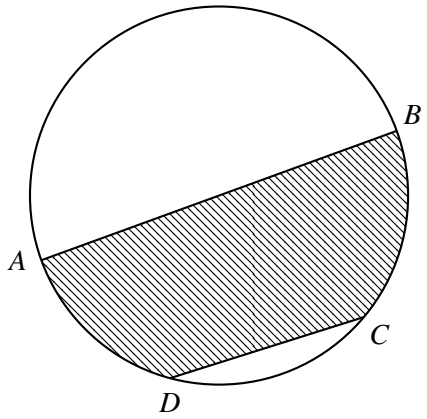
(DSE-2012 PII Q18)

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10. In the figure, $AB$ is a diameter of the circle $ABCD$. If $AB = 12$ cm and $CD = 6$ cm, then the area of the shaded region is




A. $(12\pi - 9)$ cm$^2$.

B. $(12\pi + 9)$ cm$^2$.

C. $(12\pi - 9\sqrt{3})$ cm$^2$.

D. $(12\pi + 9\sqrt{3})$ cm$^2$.

(DSE-2012 PII Q21)

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11. The rectangular coordinates of the point $P$ are $(-3, -3\sqrt{3})$. If $P$ is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(3, 150°)$.

B. $(3, 330°)$.

C. $(6, 150°)$.

D. $(6, 330°)$.

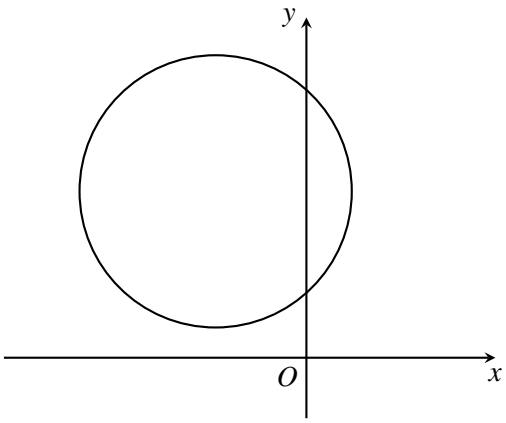
(DSE-2012 PII Q23)

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12. In the figure, the radius of the circle and the coordinates of the centre are $r$ and $(h,k)$ respectively. Which of the following are true?




I. $h+k>0$

II. $r-h>0$

III. $r-k>0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2012 PII Q26)

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13. $9 \star \diamond$ is a 3-digit number, where $\star$ and $\diamond$ are integers from 0 to 9 inclusive. Find the probability that the 3-digit number is divisible by 5.

A. $\frac{1}{5}$

B. $\frac{7}{33}$

C. $\frac{20}{99}$

D. $\frac{19}{100}$

(DSE-2012 PII Q27)

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14. The stem-and-leaf diagram below shows the distribution of the ages of a group of members in a recreational centre.

| Stem (tens) | Leaf (units) |
|---|---|
| 5 | 0 5 6 6 8 |
| 6 | 1 4 5 5 7 8 8 9 |
| 7 | 3 4 4 6 7 9 |
| 8 | |
| 9 | 1 |

A member is randomly selected from the group. Find the probability that the selected member is not under the age of 74.

A. 0.2

B. 0.3

C. 0.7

D. 0.8

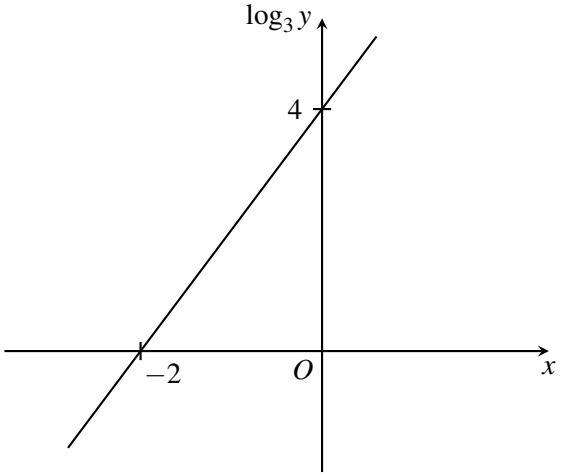
(DSE-2012 PII Q28)

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15. The graph in the figure shows the linear relation between $x$ and $\log_3 y$. If $y = mn^x$, then $n =$




A. $\frac{1}{81}$.

B. $\frac{1}{9}$.

C. 9.

D. 81.

(DSE-2012 PII Q32)

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16. $AD000002012_{16} =$

A. $(10)16^{11} + (13)16^{10} + 8210.$

B. $(10)16^{12} + (13)16^{11} + 131360.$

C. $(11)16^{11} + (14)16^{10} + 8210.$

D. $(11)16^{12} + (14)16^{11} + 131360.$

(DSE-2012 PII Q33)

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17. $i^3(3i-3) =$

A. $3+3i$.

B. $3-3i$.

C. $-3+3i$.

D. $-3-3i$.

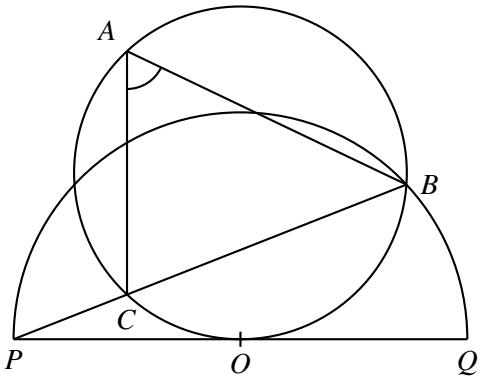
(DSE-2012 PII Q35)

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18. In the figure, $PQ$ is the tangent to the circle $ABC$ at $O$, where $O$ is the centre of the semicircle $PBQ$. It is given that $BCP$ is a straight line. If $\angle BPQ = 12°$, then $\angle BAC =$




A. 18°.

B. 24°.

C. 36°.

D. 54°.

(DSE-2012 PII Q41)

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19. The daily wage of Ada is 20% higher than that of Billy while the daily wage of Billy is 20% lower than that of Christine. It is given that the daily wage of Billy is $480.

(a) Find the daily wage of Ada.

(b) Who has the highest daily wage? Explain your answer.

(4 marks) (DSE-2012 PI Q04)

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20. There are 132 guards in an exhibition centre consisting of 6 zones. Each zone has the same number of guards. In each zone, there are 4 more female guards than male guards. Find the number of male guards in the exhibition centre. (4 marks) (DSE-2012 PI Q05)

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21. (a) Find the range of values of $x$ which satisfy both $\frac{4x+6}{7} > 2(x-3)$ and $2x-10 \leq 0$.

(b) How many positive integers satisfy both the inequalities in (a)?

(4 marks) (DSE-2012 PI Q06)

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22. In Figure 1, $AB$, $BC$, $CD$ and $AD$ are chords of the circle. $AC$ and $BD$ intersect at $E$. It is given that $BE = 8$ cm, $CE = 20$ cm and $DE = 15$ cm.

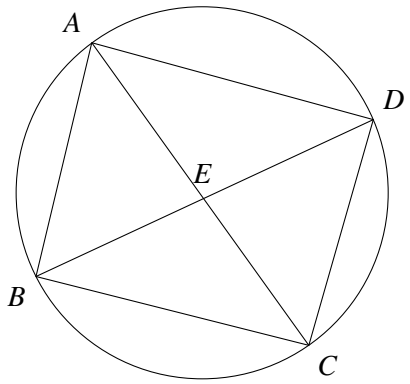



Figure 1

(a) Write down a pair of similar triangles in Figure 1. Also find $AE$.

(b) Suppose that $AB = 10$ cm. Are $AC$ and $BD$ perpendicular to each other? Explain your answer.

(5 marks) (DSE-2012 PI Q08)

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23. Tom conducts a survey on the numbers of hours spent on doing homework in a week by secondary students. Questionnaires are sent out and twenty of them are returned. The stem-and-leaf diagram below shows the numbers of hours recorded in the twenty questionnaires:

| Stem (tens) | Leaf (units) |
|---|---|
| 1 | 0 0 1 1 2 3 4 5 5 6 6 7 7 |
| 2 | 0 0 0 5 8 |
| 3 | 4 6 |

(a) Find the mean and the median of the numbers of hours recorded in the twenty questionnaires.

(b) Tom receives four more questionnaires. He finds that the mean of the numbers of hours recorded in these four questionnaires is 18. It is found that the numbers of hours recorded in two of these four questionnaires are 19 and 20.

(i) Write down the mean of the numbers of hours recorded in the twenty-four questionnaires.

(ii) Is it possible that the median of the numbers of hours recorded in the twenty-four questionnaires is the same as the median found in (a)? Explain your answer.

(6 marks) (DSE-2012 PI Q10)

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24. Let \$C be the cost of painting a can of surface area $A$ m$^2$. It is given that $C$ is the sum of two parts, one part is a constant and the other part varies as $A$. When $A = 2$, $C = 62$; when $A = 6$, $C = 74$.

(a) Find the cost of painting a can of surface area 13 m$^2$.

(b) There is a larger can which is similar to the can described in (a). If the volume of the larger can is 8 times that of the can described in (a), find the cost of painting the larger can.

(6 marks) (DSE-2012 PI Q11)

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25. (a) Figure 3(a) shows a solid metal right circular cone of base radius 48 cm and height 96 cm.

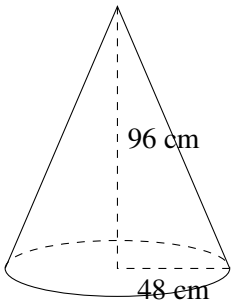



Figure 3(a)

Find the volume of the circular cone in terms of $\pi$.

(b) A hemispherical vessel of radius 60 cm is held vertically on a horizontal surface. The vessel is fully filled with milk.

(i) Find the volume of the milk in the vessel in terms of $\pi$.

(ii) The circular cone is now held vertically in the vessel as shown in Figure 3(b).

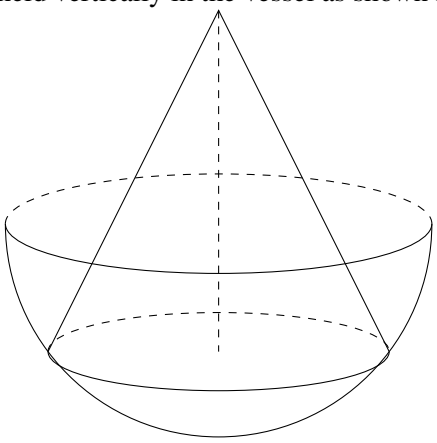


Figure 3(b)

A craftsman claims that the volume of the milk remaining in the vessel is greater than 0.3 $m^3$. Do you agree? Explain your answer.

(7 marks) (DSE-2012 PI Q12)

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26. (a) Find the value of $k$ such that $x-2$ is a factor of $kx^3-21x^2+24x-4$.

(b) Figure 4 shows the graph of $y=15x^2-63x+72$. $Q$ is a variable point on the graph in the first quadrant. $P$ and $R$ are the feet of the perpendiculars from $Q$ to the $x$-axis and the $y$-axis respectively.

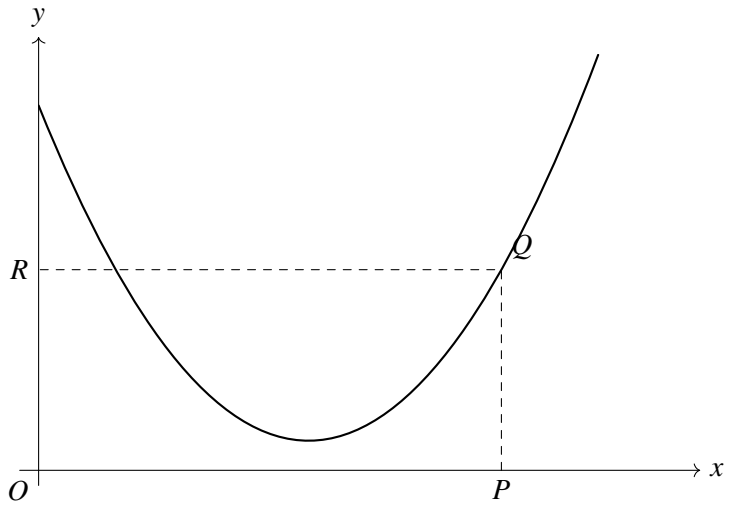



Figure 4

(i) Let $(m,0)$ be the coordinates of $P$. Express the area of the rectangle $OPQR$ in terms of $m$.

(ii) Are there three different positions of $Q$ such that the area of the rectangle $OPQR$ is 12? Explain your answer.

(7 marks) (DSE-2012 PI Q13)

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27. The standard deviation of the test scores obtained by a class of students in a Mathematics test is 10 marks. All the students fail in the test, so the test score of each student is adjusted such that each score is increased by 20% and then extra 5 marks are added.

(a) Find the standard deviation of the test scores after the score adjustment.

(b) Is there any change in the standard score of each student due to the score adjustment? Explain your answer.

(3 marks) (DSE-2012 PI Q15)

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28. The coordinates of the centre of the circle $C$ are $(6, 10)$. It is given that the $x$-axis is a tangent to $C$.

(a) Find the equation of $C$.

(b) The slope and the $y$-intercept of the straight line $L$ is $-1$ and $k$ respectively. If $L$ cuts $C$ at $A$ and $B$, express the coordinates of the mid-point of $AB$ in terms of $k$.

(7 marks) (DSE-2012 PI Q17)

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29. Figure 5(a) shows a right pyramid $VABCD$ with a square base, where $\angle VAB = 72°$. The length of a side of the base is 20 cm. Let $P$ and $Q$ be the points lying on $VA$ and $VD$ respectively such that $PQ$ is parallel to $BC$ and $\angle PBA = 60°$. A geometric model is made by cutting off the pyramid $VPBCQ$ from $VABCD$ as shown in Figure 5(b).

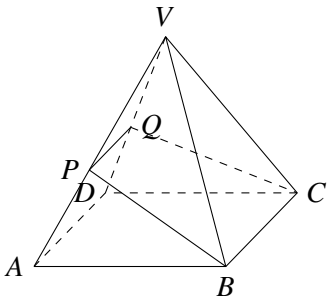



Figure 5(a)

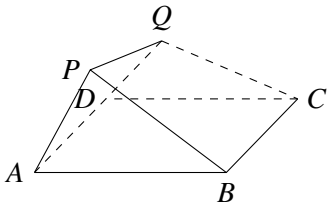



Figure 5(b)

(a) Find the length of $AP$.

(b) Let $\alpha$ be the angle between the plane $PBCQ$ and the base $ABCD$.

(i) Find $\alpha$.

(ii) Let $\beta$ be the angle between $PB$ and the base $ABCD$. Which one of $\alpha$ and $\beta$ is greater? Explain your answer.

(8 marks) (DSE-2012 PI Q18)

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30. In a city, the air cargo terminal $X$ of an airport handles goods of weight $A(n)$ tonnes in the $n$th year since the start of its operation, where $n$ is a positive integer. It is given that $A(n) = ab^{2n}$, where $a$ and $b$ are positive constants. It is found that the weights of the goods handled by $X$ in the 1st year and the 2nd year since the start of its operation are 254 100 tonnes and 307 461 tonnes respectively.

(a) (i) Find $a$ and $b$. Hence find the weight of the goods handled by $X$ in the 4th year since the start of its operation.

(ii) Express, in terms of $n$, the total weight of the goods handled by $X$ in the first $n$ years since the start of its operation.

(b) The air cargo terminal $Y$ starts to operate since $X$ has been operated for 4 years. Let $B(m)$ tonnes be the weight of the goods handled by $Y$ in the $m$th year since the start of its operation, where $m$ is a positive integer. It is given that $B(m) = 2ab^m$.

(i) The manager of the airport claims that after $Y$ has been operated, the weight of the goods handled by $Y$ is less than that handled by $X$ in each year. Do you agree? Explain your answer.

(ii) The supervisor of the airport thinks that when the total weight of the goods handled by $X$ and $Y$ since the start of the operation of $X$ exceeds 20 000 000 tonnes, new facilities should be installed to maintain the efficiency of the air cargo terminals. According to the supervisor, in which year since the start of the operation of $X$ should the new facilities be installed?

(13 marks) (DSE-2012 PI Q19)

I can finish _____ questions

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